SEM 2, 2023-24: TOPOLOGY
BACK PAPER EXAMINATION

All questions carry equal marks. Time: 3 hours. State clearly any result that you use.

(1) (a) Show that a topological space X is Hausdorff if and only if the diagonal $\Delta := \{(x,x)|x \in X\}$ is a closed subset of $X \times X$.
(b) Let $A \subset X$ and $f : A \to Y$ be a continuous function and let Y be Hausdorff. Show that if there exists a function $g : \bar{A} \to Y$ such that $g|_A = f$, then g is the unique extension of f.

(2) (a) Show that any connected open subspace of a locally path connected space is pathconnected.
(b) Is the closure of a path connected subspace path connected? Justify.

(3) (a) Prove the pasting lemma: Let $X = A \cup B$, where A and B are closed in X. $F : A \to Y$ and $g : B \to Y$ be continuous functions. If $f(x) = g(x)$ for all $x \in A \cap B$, then show that the function $h : X \to Y$ defined by $h(x) = f(x)$ if $x \in A$ and $h(x) = g(x)$ if $x \in B$ is a well defined continuous function.
(b) A map $f : X \to Y$ is said to be an *open map* if for every open set U of X, the set $f(U)$ is open in Y. Show that the projection maps $\pi_1 : X \times Y \to X$ and $\pi_2 : X \times Y \to Y$ are open maps for the product topology on $X \times Y$.

(4) Let $p : X \to Y$ be a perfect map, i.e., p is a closed continuous surjective map such that $p^{-1}(\{y\})$ is compact for each $y \in Y$.
(a) Show that if X is Hausdorff, then so is Y.
(b) Show that if X is regular, then so is Y.
(c) Show that if X is locally compact, then so is Y.

(5) (a) Prove that every compact, Hausdorff topological space is normal.
(b) Prove that a connected normal space with more than one point must be uncountable.

Date: 03-06-2024.